

Name : _____

ChE 311 Final Exam, Fall 2019

Closed book and closed notes. Only provided tables are allowed.

120 minutes, 100 points

1. (14 points) **Vector and Tensor**

For the following questions, s is a scalar, U and V are vectors, and T is a tensor. Please circle your answer.

(1) $s\nabla \cdot V$ is the same as $V \cdot \nabla s$.

(A) True (B) False

(2) $U \cdot V$ is the same as $V \cdot U$.

(A) True (B) False

(3) $\nabla \cdot VV$ is the same as $(\nabla \cdot V)V$.

(A) True (B) False

(4) UV is the same as VU .

(A) True (B) False

(5) $\nabla \cdot T$ is a vector.

(A) True (B) False

(6) ∇s is a tensor.

(A) True (B) False

(7) ∇V is a tensor.

(A) True (B) False

2. (3 points) The flow defined by the following velocity field is incompressible flow.

$$\vec{V} = (t - x - y + z)\vec{\delta}_x + (2t + x + 2y + z)\vec{\delta}_y + (3t - x - y + z)\vec{\delta}_z$$

A. True

B. False

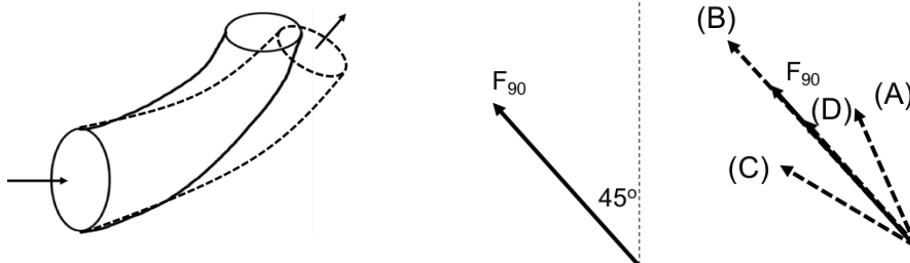
Answer: _____

3. (3 points) Under which condition(s) the initial term, $\rho \vec{v} \cdot \nabla \vec{v}$, in the N-S function can vanish as a reasonable approximation of the easier calculation.

(A) Low Reynolds number, creeping flow.
 (B) High Reynolds number, inviscid flow.
 (C) Unidirectional flow, rectilinear motion.
 (D) Turbulent flow.

Answer: _____

4. (5 points) For a pipe bend, if we change the bending angle from 90° to 60° , please choose the one best represents the force applied on the 60° pipe bend to balance the force by the fluid on the pipe. The force previously applied on the 90° pipe bend is the solid arrow on the figure.



Answer: _____

5. (5 points) For a 90° pipe bend, the pipe diameter is 20 cm. The flow rate is 100 L/min. Please estimate the Reynolds number for water flow through the pipe. Consider water density 1000 kg/m^3 and viscosity 1 cp at room temperature.

(A) 10 (B) 10^2 (C) 10^3 (D) 10^4 (E) 10^5

Answer: _____

6. (20 points) **Using a spherical funnel to measure the viscosity of a fluid.**

Using spherical funnel to measure the viscosity of wine. The funnel was fully filled up initially. The draining time (t_{efflux}) was measured 100 seconds. The radius of the tank is 5 cm. The radius of the pipe is 1 mm and the length is 10 cm. The density of the wine in room temperature is about 1000 kg/m^3 . Please calculate the viscosity of the wine in this case.

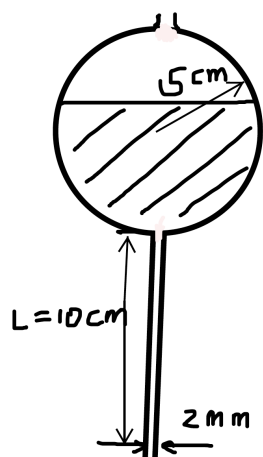


Figure 1. Dimensions of the spherical funnel.

7. (20 points) **Turbulent flow in a circular tube.**

(a) What is Reynolds stress for $\tau_{rz}^{(t)}$?

(b) Use eddy viscosity model to treat the above Reynolds stress for $\tau_{rz}^{(t)}$. Please write down the expression.

(c) Write down the total stress τ_{rz} , including the viscous stress and Reynolds stress.

(d) Use the following number to evaluate $\mu^{(t)}/\mu$ at $y = R/2$.

$R = 8\text{cm}$

Wall shear stress, $\tau_0 = 0.15\text{ Pa}$

Density of the fluid, $\rho = 1000\text{ kg/m}^3$.

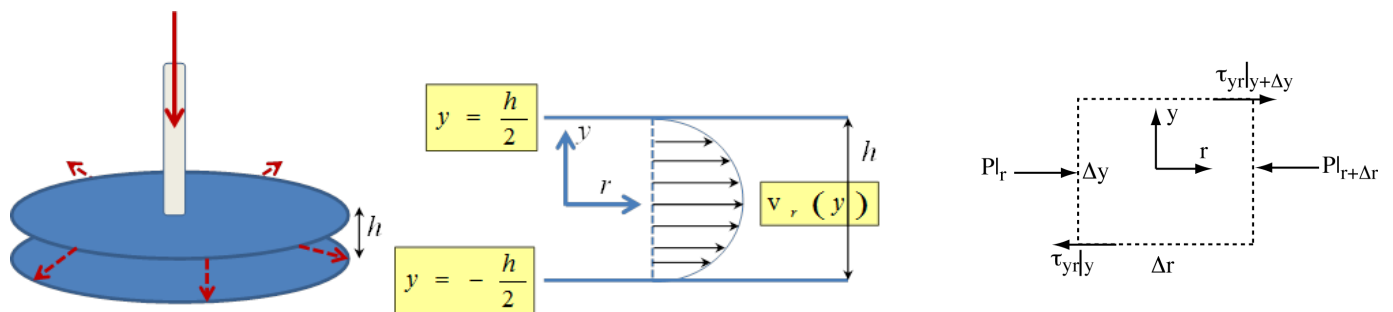
Viscosity of the fluid, $\nu = 10^{-6}\text{ m}^2/\text{s}$.

For viscous sublayer, $\frac{\bar{v}_x}{v_*} = \frac{yv_*}{\nu} \left[1 - \frac{1}{2} \left(\frac{\nu}{v_* B} \right) \left(\frac{yv_*}{\nu} \right) - \frac{1}{4} \left(\frac{yv_*}{14.5\nu} \right)^3 + \dots \right] \quad 0 < \frac{yv_*}{\nu} < 5$

For inertial sublayer, $\frac{\bar{v}_x}{v_*} = 2.5 \ln \left(\frac{yv_*}{\nu} \right) + 5.5 \quad \frac{yv_*}{\nu} > 30$

8. (30 points) **Flow between two parallel plates**

A radial-flow channel has been developed to measure the adhesion (energy) of cells to surfaces *in vitro*. The radial-flow channel is shown in the figure below. In this device, fluid flows in from a central source, flows out in a radial direction and the volumetric flow rate at any positions r is Q . The flow itself is driven by an unspecified pressure gradient.



(a) Solve for the velocity profile.

- (b) Use your solution for the velocity profile and the constant flow rate condition to derive the wall shear stress at $y = h/2$ as a function of μ , Q , r , h .